

Advanced (Carolina Reaper)

- Q1.** What is the equation of the line in point-slope form that passes through the points $(-2, 3)$ and $(1, 5)$?
- (A) $y - 3 = \frac{2}{3}(x + 2)$
 (B) $y - 5 = \frac{2}{3}(x - 1)$
 (C) $y - 3 = \frac{1}{2}(x + 2)$
 (D) $y - 5 = \frac{1}{2}(x - 1)$
 (E) $y - 3 = \frac{3}{2}(x + 2)$
- Q2.** Which equation represents a line with a slope of -2 and a y -intercept of 4 ?
- (A) $y = -2x - 4$
 (B) $y = -2x + 4$
 (C) $y = 2x - 4$
 (D) $y = 2x + 4$
 (E) $y = -2x + 2$
- Q3.** A bakery sells a total of 250 loaves of bread per day. The daily profit is given by the equation $P = 2x - 50$, where x is the number of loaves sold. What is the profit if the bakery sells 150 loaves of whole wheat bread at _____

 each and 100 loaves of white bread at _____

 each?
- (A) $P = 200$
 (B) $P = 250$
 (C) $P = 300$
 (D) $P = 350$
 (E) $P = 400$
- Q4.** The standard form of the equation of a line is $Ax + By = C$. If $A = 2$, $B = 3$, and $C = 6$, what is the slope of the line?
- (A) $m = -\frac{2}{3}$
 (B) $m = -\frac{3}{2}$
 (C) $m = \frac{2}{3}$
 (D) $m = \frac{3}{2}$
 (E) $m = 1$
- Q5.** The equation $y = 3x - 2$ represents a line with a slope of 3 and a y -intercept of -2 . What is the x -intercept of this line?
- (A) $x = \frac{2}{3}$
 (B) $x = -\frac{2}{3}$
 (C) $x = \frac{3}{2}$
 (D) $x = -\frac{3}{2}$
 (E) $x = 0$
- Q6.** Tom has been saving money for a new bike and has 120 dollars in his piggy bank. He wants to buy a bike that costs 180 dollars. If he saves 5 dollars per week, how many weeks will it take him to have enough money to buy the bike?
- (A) 6 weeks
 (B) 8 weeks
 (C) 10 weeks
 (D) 12 weeks
 (E) 15 weeks
- Q7.** A water tank can hold 1200 gallons of water. If 300 gallons of water are already in the tank, and 50 gallons of water are added every hour, how many hours will it take to fill the tank?
- (A) 12 hours
 (B) 15 hours
 (C) 18 hours
 (D) 20 hours
 (E) 22 hours
- Q8.** The point-slope form of a line is $y - 2 = 3(x - 1)$. What is the slope-intercept form of this line?
- (A) $y = 3x - 1$
 (B) $y = 3x + 1$
 (C) $y = 3x - 3$
 (D) $y = 3x + 3$
 (E) $y = 2x + 1$

Open Ended Questions (FRQ)

- Q9.** The cost of producing x units of a product is given by the equation $C = 2x + 500$. The revenue from selling x units is given by the equation $R = 5x$. Find the break-even point, where the revenue equals the cost. Show all work and explain your answer.
- Q10.** A person invests _____

 2000
 in a savings account that earns an annual interest rate of 5%. The interest is compounded annually. How much will the person have in the account after 10 years? Use the equation $A = P(1 + r)^n$, where A is the amount of money accumulated after n years, including interest, P is the principal amount (initial investment), r is the annual interest rate (in decimal), and n is the number of years.